

Muhammad Subhan

Python & AI Automation Engineer

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SUMMARY

Python & AI Automation Engineer with **1+ year of hands-on, production experience** in FastAPI/Django backend development, biometric hardware integration (ZKTeco), Retrieval-Augmented Generation (RAG) systems, and workflow automation with n8n. Over this time, has designed and shipped a desktop biometric attendance system used by real clients, built an AI-powered automated exam grading pipeline for a professional accountancy college, built multiple secure REST APIs, and delivered a suite of production-grade RAG chatbots and n8n automation pipelines. Skilled in building REST APIs, AI chatbots, and LLM-based applications using LangChain, LangGraph, and vector databases for real-world environments, with a strong focus on reliability, clean architecture, and direct client support.

TECHNICAL SKILLS

- **Backend:** Python, FastAPI, Django, Flask, REST API Development, API Integration
- **AI / LLM:** Retrieval-Augmented Generation (RAG), LangChain, LangGraph, AI Agents, Vision-based Document Extraction
- **Automation:** n8n Workflow Automation, Webhooks, AI Agent Pipelines, Python Scripting
- **Biometrics:** ZKTeco Integration — Fingerprint, Face Recognition, Palm, RFID Card Readers
- **Databases:** SQL, NoSQL, MongoDB, MySQL, PostgreSQL, SQLite3, Redis, Vector Databases (Qdrant)
- **Frontend:** HTML, CSS, Tailwind CSS, React.js, Next.js
- **DevOps:** Docker, Kubernetes
- **Tools:** Git, GitHub, Postman, VS Code, Windsurf

WORK EXPERIENCE

AI Developer - PAC College of Accountancy

September 2026 — Present

- Designed and built an AI-powered automated exam-grading pipeline that extracts marking schemes and student answer scripts from scanned PDFs using vision-based LLM extraction, grades them against structured rubrics, and returns an annotated, marked-up PDF with a verifiable CSV breakdown.
- Engineered a MongoDB-backed data pipeline (model answers, student submissions, and grade documents) with a reproducible two-pass rubric-extraction workflow, including an own-figure (OF) dependency graph so students receive correct method-mark credit even when an earlier figure in their working is wrong.
- Built a frontend for markers to upload a student script, select the saved marking scheme, and download the graded, annotated PDF alongside a per-criterion CSV for audit.
- Implemented deterministic post-processing guards around the LLM grading step (duplicate-credit prevention, sign verification, evidence-column disambiguation) so the saved mark, not the raw model reply, is the source of truth.

Full Stack Developer - Mozzine Technologies (On-site)

March 2025 — April 2026

- Built a desktop biometric attendance application using Python and PyQt6 (MVC architecture), integrating ZKTeco devices with SQLite3 for attendance data processing.
- Developed secure FastAPI REST APIs for device communication, authentication, and data synchronization.
- Integrated multiple ZKTeco device models — fingerprint, face recognition, palm recognition, and RFID card readers — and implemented multi-threaded sync workers (QThread) to keep the UI responsive.
- Provided direct customer support for the full one-year engagement: diagnosed and resolved recognition errors, connectivity, enrollment, sync failures, and deployment issues, while directly handling day-to-day client queries and escalations alongside the biometric system delivery.
- Created n8n workflow automations to streamline operational tasks and orchestrate data pipelines.

PROJECTS

AI RAG Agents & Chatbots (Python, FastAPI, Claude AI, n8n, Vector DB)

Customer Support RAG Agent

- Built an automated query-resolution agent that combines an LLM with vector search over a curated knowledge base, cutting manual ticket handling for repetitive questions.
- Triggered and orchestrated through n8n pipelines, allowing new documents and FAQs to be ingested and indexed without redeploying the backend.
- Tuned retrieval parameters (chunk size, top-k, embedding model) to balance answer relevance with response latency.

Document Q&A RAG Agent

- Enabled users to upload PDFs and other documents and query them in natural language for fast policy and manual lookups, removing the need to manually search long documents.
- Implemented chunking and embedding pipelines with a vector database to keep retrieval accurate across large, multi-page files.
- Extended the same architecture into a dedicated Product Knowledge RAG Bot for pre-sales and technical queries, giving sales and support teams instant, source-grounded answers.

Multi-Source RAG Agent

- Designed an agent that aggregates data from multiple databases, external APIs, and documents to answer complex business queries in real time.
- Used LangChain/LangGraph to orchestrate tool calls across sources and merge results into a single coherent, cited response.
- Handled edge cases such as conflicting data between sources and missing records, improving reliability of business-critical answers.

Domain-Specific RAG Chatbots

- Built a Medical & Healthcare Info Bot and a CSS/PPSC Exam Preparation Agent, each with topic-wise Q&A over curated subject-matter content.
- Developed general-purpose chatbots with persistent conversation memory, deployed on both web and WhatsApp for wider accessibility.
- Applied prompt engineering and context-window management to keep multi-turn conversations coherent and on-topic across sessions.

n8n Workflow Automation (n8n, Python, REST APIs, Webhooks)

Lead Capture & Data Sync

- Automated the flow from form submissions to CRM entry, including auto-generated welcome emails to new leads.
- Built scheduled and webhook-triggered syncs between Google Sheets, Airtable, and PostgreSQL to keep lead data consistent across systems.
- Reduced manual data entry and eliminated sync delays between marketing tools and the sales database.

Notification & Alert System

- Built workflows that continuously monitor API endpoints for uptime and performance.
- Configured real-time Slack and email alerts triggered on failures or threshold breaches, reducing incident response time.
- Added retry and escalation logic so critical failures were surfaced to the right channel without flooding the team with noise.

EDUCATION

BS Computer Science - University of Narowal (UON)

Nov 2021 - May 2025

Intermediate (Pre-Engineering) - Royal College of Science, Narowal

May 2018 - Oct 2020

OTHER SKILLS AND EXPERIENCE

Teacher - Al-Karam Scholars School, Sadowala Uncha, Narowal - Matric Level (9th & 10th Grade)

3.5 Years

- Delivered classroom lectures to 9th and 10th grade (Matric level) students for 3.5 years.
- Prepared topic-wise study material and simplified complex concepts to match each student's understanding and pace.
- Mentored students on exam preparation, assignments, and problem-solving techniques, helping several improve their board exam performance.
- Developed strong communication, patience, and explanation skills by regularly breaking down technical topics for young learners.

Online Home Tutor - Matric Level (9th & 10th Grade)

1 Year

- Delivered live online classes to 9th and 10th grade students via Zoom and Google Meet, adapting classroom teaching methods to a fully remote format.
- Conducted one-on-one virtual tutoring and doubt-clearing sessions outside standard class hours.
- Maintained consistent student engagement and progress tracking in an online learning environment.

Soft Skills

- **Communication:** Able to clearly explain technical concepts to both technical and non-technical stakeholders, including clients and students.
- **Problem-Solving:** Strong analytical and troubleshooting ability, developed through diagnosing hardware, software, and integration issues for clients.
- **Teamwork & Collaboration:** Comfortable working across cross-functional teams and coordinating with clients, support staff, and developers.
- **Time Management:** Skilled at balancing multiple projects, deadlines, and support requests simultaneously without compromising quality.
- **Adaptability:** Quick to learn new tools, frameworks, and client requirements in fast-changing project environments.
- **Client Handling & Patience:** Experienced in direct client support and teaching, requiring patience, empathy, and clear step-by-step guidance.
- **Leadership & Mentoring:** Comfortable guiding students and junior team members, gained through 3.5 years of teaching experience.
- **Work Ethic & Reliability:** Dependable in meeting commitments, following through on client issues, and maintaining consistent quality of output.